

Eigenvalue Patterns for Common Matrix Types

Concise exam revision table for recognizing eigenvalue behavior quickly.

Based on handwritten MIT 18.06 revision notes.

1 Core Idea

Core idea. Special matrix structure often predicts eigenvalue behavior before calculation. Use structure first, then verify by computation.

This note is an original compact reference based on standard 18.06 facts.

2 Common Patterns

Matrix type	Eigenvalue pattern	Exam use
Symmetric $S = S^T$	Real eigenvalues.	Orthogonal eigenvectors; use $S = Q\Lambda Q^T$.
Orthogonal $Q^T Q = I$	$ \lambda = 1$.	Lengths are preserved.
Skew-symmetric $A^T = -A$	Pure imaginary or zero eigenvalues.	Diagonal entries are zero.
Projection $P^2 = P$	$\lambda = 0$ or 1 .	Range stays, nullspace collapses.
Positive definite S	All $\lambda > 0$.	Use for quadratic forms and pivots.
Rank-one uv^T	At most one nonzero eigenvalue.	Most directions collapse.
Triangular	Diagonal entries.	Read eigenvalues directly.
Similar matrices	Same eigenvalues.	Change coordinates without changing spectrum.

3 Shift, Inverse, and Powers

If $Ax = \lambda x$, then:

$$(A + cI)x = (\lambda + c)x.$$

So shifting by cI shifts eigenvalues by c .

If A is invertible, then:

$$A^{-1}x = \frac{1}{\lambda}x.$$

So inverse eigenvalues are reciprocals.

For powers:

$$A^k x = \lambda^k x.$$

This is why eigenvalues control long-term behavior.

4 Stability Memory

Problem type	Stability test
Repeated powers A^k	Use $ \lambda $.
Differential equation $u' = Au$	Use $\operatorname{Re} \lambda$.
Projection	Stable because eigenvalues are 0, 1.
Orthogonal iteration	Lengths do not grow because $ \lambda = 1$.

5 Common Mistakes

Mistake	Correct exam habit
Using $S = Q\Lambda Q^T$ without symmetry.	First check $S = S^T$.
Assuming all real matrices have real eigenvalues.	Rotations can give complex eigenvalues.
Forgetting zero eigenvalues in projections.	Projection kills directions in the nullspace.
Using $ \lambda $ for $u' = Au$.	Use $\operatorname{Re} \lambda$ for continuous systems.
Assuming same eigenvalues means same eigenvectors.	Similar matrices preserve eigenvalues, not necessarily vector coordinates.

6 Final Exam Checklist

Checklist.

1. Identify special structure before calculating.
2. Check symmetry, orthogonality, projection, or triangular form.
3. Use the structure to predict eigenvalue behavior.
4. Verify with trace, determinant, or $\det(A - \lambda I)$.
5. Use shifts, inverses, and powers to transform known eigenvalues.
6. Use $|\lambda|$ for powers and $\operatorname{Re} \lambda$ for differential equations.